Ganesh Reddy
Gerhart-Hauptmann-Str. 11
91058 Erlangen, Germany
+49 1551 0182395
ganeshreddy30102000@gmail.com

January 28, 2026

Mischa Dombrowski
Friedrich-Alexander-Universität Erlangen–Nürnberg
Erlangen, Germany

Subject: Application for Master's Project thesis – Neonatal Chest X-ray Generation in the Low Data Regime

Dear Mr. Dombrowski,

I am writing to express my strong interest in the Master's thesis topic *"Neonatal Chest X-ray Generation in the Low Data Regime."* I am currently pursuing a Master of Science in Autonomy Technologies at Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU), with a strong academic and practical focus on medical image analysis, computer vision, and deep learning. The opportunity to work with real clinical data in a hospital environment and address data scarcity in neonatal imaging strongly aligns with my academic background and research interests.

During my studies and projects, I have gained hands-on experience working with chest X-ray images for disease classification tasks, where I trained convolutional neural networks using public medical imaging datasets. I applied transfer learning and data augmentation techniques to improve model performance under limited data conditions and evaluated robustness across patient distributions. Through this work, I developed a strong understanding of dataset bias, domain shift, and the challenges of generalizing models trained on adult datasets to new or unseen domains.

In addition, I currently work as a Student Research Assistant at FAU, where I develop and maintain Python-based data analysis pipelines, support data validation, and create automated evaluation scripts for scientific experiments. This role has strengthened my ability to work independently, conduct systematic experimentation, and document results in a structured and reproducible manner—skills that are essential for research in clinical environments with limited compute resources.

I am particularly motivated by the second part of the thesis, which explores synthesizing medical images from vital parameters. The idea of reducing radiation exposure by leveraging learned image representations and generative models resonates strongly with my interest in clinically impactful AI. I am eager to explore approaches involving supervised learning, domain adaptation, and foundation models trained on large public datasets, while carefully evaluating their limitations in neonatal settings.

I am highly motivated, curious, and comfortable working at the intersection of engineering and clinical research. I would welcome the opportunity to contribute to this project and further develop my expertise in medical image computing under real-world constraints.

Thank you very much for your time and consideration. I would be happy to discuss my background and motivation in more detail.

Sincerely,
Ganesh Reddy